	Program	**B. Tech Computer Science and Engineering**				**Program Code**	**24BTHCSE**	
	Course	**Fundamentals of AI and ML**				**Course Code**	**24BTPHY106**	
	Semester	**I**	**Credits**	**3**	**Theory** / **Practical**	**Total Hours**	**40**	
					3 / **0**			

	COURSE OBJECTIVES:
1) 2) 3) 4)	Study the concepts of Artificial Intelligence. Learn the methods of solving problems using Artificial Intelligence. Learn about Machine learning and NLP concepts
	General Instructions for Teaching-Learning:
1) 2) 3) 4)	PPT, Creative thinking, Flipped Learning, Problem Based learning Enhance students' knowledge with statistical and mathematical concepts Provide opportunities for hands-on experience with AI and ML tools and techniques. Emphasize project-based learning to provide students with practical experience in developing AI and ML

Module	**Topics**	**Hours**
I	**Introduction and Intelligent Agents** What Is AI? — The Foundations of Artificial Intelligence — The History of Artificial Intelligence — The State of the Art — Risks and Benefits of AI Agents and Environments — Good Behavior: The Concept of Rationality — The Nature of Environments — The Structure of Agents	8
II	**Problem-solving - Solving Problems by Searching** Solving Problems by Searching Problem-Solving Agents — Example Problems — Search Algorithms — Uninformed Search Strategies — Informed (Heuristic) Search Strategies — Heuristic Functions	8
III	**Knowledge, reasoning, and planning** **Logical Agents** - Knowledge-Based Agents — The Wumpus World — Logic — Propositional Logic: A Very Simple Logic — Propositional Theorem Proving — Effective Propositional Model Checking — Agents Based on **Propositional Logic** First-Order Logic - Representation Revisited — Syntax and Semantics of First-Order Logic — Using First-Order Logic — Knowledge Engineering in First-Order Logic **Inference in First-Order Logic** Propositional vs. First-Order Inference — Unification and First-Order Inference — Forward Chaining — Backward Chaining — Resolution	8
IV	**Machine Learning** Learning from Examples — Forms of Learning — Supervised Learning — Learning Decision Trees — Model Selection and Optimization — The Theory of Learning — Linear Regression and Classification — Nonparametric Models — Ensemble Learning — Developing Machine Learning Systems	8
V	**NATURAL LANGUAGE PROCESSING** Language Models - Grammar - Parsing - Augmented Grammars - Complications of Real Natural Language - Natural Language Tasks	8

COURSE OUTCOMES:
At the end of Course Students will be, **CO1:** Familiar with Artificial Intelligence, its foundation and principles.

CO2: Identify appropriate AI methods to solve a given problem.
CO3: Study about Logical agents, reasoning and planning.
CO4: Understand the fundamental concepts of Machine Learning.
CO5: Understand the application of Natural language processing.

Text Books:

1. Artificial Intelligence: A Modern Approach, 4th edition, Published by Pearson 2021 Stuart Russell , Peter Norvig
2. Elaine Rich, Kevin Knight, Shivashankar B Nair," Artificial Intelligence", 3rd Edition, Tata Mc Graw Hill, 2013.

Reference Books:

1. Artificial Intelligence Structure and strategies for complex problem solving George F Lugar 5th Edition Pearson 2011 Jean-Louis Ermine," Expert Systems: Theory and Practice", Prentice Hall of India, 2004
2. E. Charnaik and D. McDermott, " Introduction to artificial Intelligence Pearson Education, 2012.
3. Kevin Night and Elaine Rich, Nair B., "Artificial Intelligence (SIE)", Mc Graw Hill2008.

E-Learning Sources:

1. https://www.coursera.org/courses?query=artificial%20intelligence
2. https://www.javatpoint.com/first-order-logic-in-artificial-intelligence
3. https://www.geeksforgeeks.org/artificial-intelligence-an-introduction/